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Fuel-Optimal Trajectory Planning for Lunar Vertical Landing

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In this paper we consider a trajectory planning problem arising from a lunar vertical landing with minimum fuel consumption. The vertical landing requirement is written as a final steering angle constraint, and a nonnegative regularization term is proposed to modify the cost functional. In this way, the final steering angle constraint will be inherently satisfied according to Pontryagin's Minimum Principle. As a result, the modified optimal steering angle has to be determined by solving a transcendental equation. To this end, a transforming procedure is employed, which allows for finding the desired optimal steering angle by a simple bisection method. Consequently, the vertical landing trajectory can be found offline by the indirect shooting method. Finally, some numerical examples are presented to demonstrate and verify the developments of the paper.

Keywords: Lunar exploration; vertical soft landing; final control constraint; trajectory optimization; indirect shooting method.

1. Introduction

As the nearest celestial body to the Earth, the Moon has been increasingly explorated recently.[1] Soft landing is the key to deploying the lunar lander to the lunar surface for scientific missions. The typical soft landing consists of two phases, i.e. de-orbit burn phase and powered descent phase.[2] During the de-orbit burn phase, the altitude of the lunar lander is reduced from a parking orbit to the perilune altitude. The powered descent phase is expected to use the retro-propulsion provided by the engine to meet the required final condition, either at the landing site or above the landing site.[3] The well-known Apollo lunar descent guidance represented the thrust acceleration vector as a linear function of time.[4] However, the upper and lower bounds on the

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thrust magnitude were not considered, which may lead to infeasible thrust acceleration vector command. In addition, it has been reported that the Apollo lunar descent guidance is not optimal with respect to fuel consumption.[5,6] In fact, fuel consumption plays a significant role in space mission designs. Fuel-optimal trajectories enable the spacecraft to carry heavier payloads or extend mission durations by maximizing the efficiency of the propulsion system. This opens up opportunities for more extensive scientific experiments, technology demonstrations, and resource utilization on the lunar surface. Thus, significant efforts have been devoted to designing the fuel-optimal landing trajectory, which is equivalent to solving an Optimal Control Problem (OCP).

Due to the inherent nonlinearity of many real-world applications, numerical methods are commonly required to solve OCPs. An OCP can be numerically solved via the direct or the indirect method, as summarized in Ref. 7. The direct method discretizes the control and/or the state variables and parameterizes the OCP into a Nonlinear Programming (NLP) problem.[8] The resulting NLP problem can be solved by the interior-point or the sequential quadratic programming methods. The direct method, especially convex optimization based one, has been widely used for onboard implementation for soft landing problems; see Refs. 9–11. On the other hand, the indirect method employs Pontryagin's Minimum Principle (PMP) to transform the OCP into a Two-Point Boundary-Value Problem (TPBVP), which is then usually solved by the indirect shooting method. Albeit not ideal for onboard implementation, this method has been widely used to reveal the structure of the optimal control law; see Refs. 3,12,13. In the light of both direct and indirect methods suffering from the convergence issue and the computational burden,[14] machine learning-based methods have been frequently used in aerospace engineering. Generally, there are two quite different machine learning-based approaches. The first class, known as reinforcement learning, involves training an agent to behave in a potentially changing environment via repeated interactions, and the goal is to maximize a carefully designed reward function.[15–17] In contrast to reinforcement learning, behavioral cloning aims to clone the observed expert behavior through training a neural network on optimal state–action pairs; see Refs. 18–24.

Most of these methods mentioned in the preceding paragraph, whether the obtained solution is real time or not, have solely focused on OCPs without considering the final control constraint. From a practical point of view, the final control constraint can be imposed to guarantee smooth system operation,[25] reduce final errors,[26] or improve the missile's guidance performance,[27–30] to name a few. For the lunar landing problem, the thrust engine is *de facto* usually fixed with the lunar lander's body. Therefore, the steering angle essentially represents the attitude of the lunar lander.[13] A vertical landing, characterized by a final steering angle constraint, can prevent the lunar lander from rollover, or be further used for detecting obstructions and implementing terminal avoidance manoeuvres.[31] For this reason, there were some works to generate the fuel-optimal vertical landing trajectory; see Refs. 2, 3, 31–34. Specifically, in Ref. 2, the lunar landing trajectory was deconstructed into two parts, and the final steering angle constraint was augmented into the Hamiltonian along the second part. Then, the original TPBVP became a Multi-PBVP. This would inevitably complexify the solving

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procedure. In Ref. 3, an improved estimation of the time-to-go from Ref. 35 was implemented into the Apollo guidance proposed in Refs. 4, 36, and the vertical landing was realized at the cost of using extra fuel. This approach added the terminal descent phase into the powered descent phase, and an accurate estimation of the time-to-go was needed in order to derive the solution close to optimal. Meanwhile, to avoid the singularity problem caused by the touchdown point when the time-to-go reaches zero, a constant thrust acceleration value was obtained by solving a quadratic equation. In this way, the resulting thrust acceleration was used shortly before touchdown. In Ref. 32, a control parameterization and time scaling transform technique were used to realize a soft landing with the terminal attitude of the lunar lander to be within a small deviation from vertical landing. In Ref. 33, to deal with the final steering angle constraint, a time varying matrix was introduced to the cost functional, and the resulting problem was solved via model predictive static programming. However, the nonanalytical time varying matrix may not be generalizable to broader cases. Meanwhile, the time varying matrix needs to be determined using the iteration procedure, which may further complexify the solving procedure. In Ref. 34, a nonnegative regularization term was proposed to augment the cost functional, ensuring that the final steering angle constraint is met. Nevertheless, in that work, the thrust magnitude of the lunar lander is not adjustable.

In this paper, to cope with the final steering angle constraint, which can be seen as a final control constraint, we modify the cost functional by augmenting a simple nonnegative regularization term, as done in Ref. 34. In this way, the final steering angle constraint will be satisfied at the final time automatically because PMP minimizes the augmented cost functional. The proposed method can be readily embedded into the indirect shooting method once the modified optimal steering angle is obtained by solving a transcendental equation, whose desired zero can be found by a simple bisection method. Unlike the existing research,[2,3,31–33] there is no need for adding the terminal descent phase and estimating the time-to-go. Meanwhile, the proposed method is easy-to-implement and requires minimal parameter settings. More importantly, different from the thrust magnitude solution in Refs. 3 and 33, the thrust magnitude obtained by the proposed method is in a bang-bang form, which theoretically guarantees the solution optimality.

The remainder of the paper is organized as follows. Section 2 presents the OCP regarding the lunar vertical landing. PMP is used to derive the necessary conditions for optimality without and with the final steering angle constraint in Sec. 3. Numerical simulations are presented in Sec. 4, showing that the proposed method is able to find the fuel-optimal landing trajectory with only negligible penalty on the fuel consumption, and the final steering angle constraint ensuring a vertical landing can be strictly satisfied. Section 5 concludes.

2. Problem Formulation

After the de-orbit burn phase, i.e. reducing the altitude of the lunar lander from the parking orbit to the perilune altitude, is completed, the powered descent phase can

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be initiated according to the powered descent initiation condition.[3] However, to achieve the vertical landing, it is usually needed to include the terminal descent phase. In this paper we solve the vertical landing problem without the terminal descent phase. In this way, the algorithm applies to the entire flight until touchdown once the powered descent phase is initiated.

Consider the 2-dimensional lunar soft landing problem in Ref. 37, as shown in Fig. 1, where a flat nonrotating Moon model with constant gravity is assumed. The origin of the coordinate system Oyz is located at the landing site, with $y \in \mathbb{R}$ representing the lunar lander's ground range and $z \geq 0$ denoting its altitude. The lunar lander is propelled by an engine with adjustable thrust magnitude and steering angle. The thrust magnitude is denoted by $T \in [0, T_{\mathrm{m}}]$, where the constant T_{m} means the maximum thrust magnitude. Let $u \in [0, 1]$ denote the engine thrust ratio, then we have $T = uT_{\mathrm{m}}$. The steering angle, denoted by $\theta \in [-\pi, \pi]$, is defined to be the angle from the thrust vector to the local vertical line of the lunar lander. In addition, the gravity of the lunar lander is mg, in which m is its mass, and g is the Moon's gravitational acceleration. Denote by $\boldsymbol{r} = [y; z]$ the position vector, $\boldsymbol{v} = [v_y; v_z]$ the velocity vector of the lunar lander, respectively; v_y is the horizontal speed and v_z is the vertical speed. Let $\boldsymbol{x} = [\boldsymbol{r}, \boldsymbol{v}, m]^T$ be the state vector. Then, the dynamics of the lunar lander is represented by

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{f}(\boldsymbol{x}, u, \theta, t) \Rightarrow \begin{cases} \dot{\boldsymbol{r}}(t) = \boldsymbol{v}(t), \\ \dot{\boldsymbol{v}}(t) = \boldsymbol{g} + \dfrac{u(t)T_{\mathrm{m}}}{m(t)}\hat{\boldsymbol{i}}_\theta(t), \\ \dot{m}(t) = -\dfrac{u(t)T_{\mathrm{m}}}{I_{\mathrm{sp}}g_0}, \end{cases} \tag{1}$$

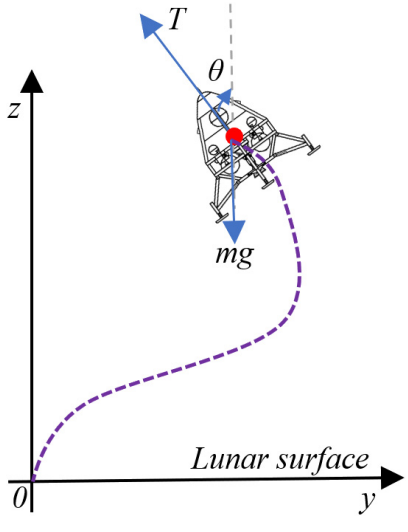


Fig. 1. Coordinate system for the lunar vertical landing.

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where $t \geq 0$ is the time; $\boldsymbol{f} : \mathbb{R}^5 \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}_0^+ \to \mathbb{R}^5$ is the smooth vector field. The acceleration vector $\boldsymbol{g} = [0; -g]$. $\hat{\boldsymbol{i}}_\theta = [\sin\theta; \cos\theta]$ is the unit thrust vector. The constant I_{sp} represents the engine's specific impulse and g_0 is the Earth's gravitational acceleration at sea level. Note that this dynamical model applies to other landing problems when the planet is assumed to have no atmosphere or the atmosphere is sufficiently thin to be ignorable in the powered descent phase.[3]

The initial condition of the lunar lander at $t_0 = 0$ is given by

$$\boldsymbol{r}(0) = \boldsymbol{r}_0, \quad \boldsymbol{v}(0) = \boldsymbol{v}_0, \quad m(0) = m_0. \tag{2}$$

The final condition at touchdown is

$$\boldsymbol{r}(t_f) = \boldsymbol{0}, \quad \boldsymbol{v}(t_f) = \boldsymbol{0}, \tag{3}$$

where t_f is the free final time. Because the steering angle defines the body attitude of the lunar lander,[13] to ensure a vertical landing, the final steering angle should satisfy the following constraint:

$$\theta(t_f) = 0. \tag{4}$$

The cost functional to be minimized is the fuel consumption during landing, which is equivalent to minimizing the following cost functional:

$$J = \int_0^{t_f} u(t)\mathrm{d}t. \tag{5}$$

3. Fuel-Optimal Trajectory Planning

In this section, we first present the indirect method to design the fuel-optimal landing trajectory based on PMP. Then, we shall show an easy-to-implement approach to deal with the final steering angle constraint by simply augmenting a nonnegative term into the original cost functional.

3.1. *Without the final steering angle constraint*

Denote by $\boldsymbol{p}_r = [p_y; p_z]$, $\boldsymbol{p}_v = [p_{v_y}; p_{v_z}]$, and p_m the co-state variables of $\boldsymbol{r}$, $\boldsymbol{v}$, and m, respectively. Then, the Hamiltonian is expressed as follows:

$$\mathscr{H} = \boldsymbol{p}_r^T \cdot \boldsymbol{v} + \boldsymbol{p}_v^T \cdot \left(\boldsymbol{g} + \frac{uT_{\text{m}}}{m}\hat{\boldsymbol{i}}_\theta\right) + p_m\left(-\frac{uT_{\text{m}}}{I_{\text{sp}}g_0}\right) + u. \tag{6}$$

According to PMP,[38] it holds that

$$\begin{cases} \dot{\boldsymbol{p}}_r(t) = -\dfrac{\partial\mathscr{H}}{\partial\boldsymbol{r}(t)} = \boldsymbol{0}, \\ \dot{\boldsymbol{p}}_v(t) = -\dfrac{\partial\mathscr{H}}{\partial\boldsymbol{v}(t)} = -\boldsymbol{p}_r(t), \\ \dot{p}_m(t) = -\dfrac{\partial\mathscr{H}}{\partial m(t)} = \dfrac{u(t)T_m}{m^2(t)}\boldsymbol{p}_v^T(t) \cdot \hat{\boldsymbol{i}}_\theta(t). \end{cases} \tag{7}$$

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The optimal steering angle follows that

$$\hat{\boldsymbol{i}}_{\theta}^{*}(t) = -\frac{\boldsymbol{p}_v(t)}{\|\boldsymbol{p}_v(t)\|_2}. \tag{8}$$

It is shown in Ref. 3 that the optimal thrust magnitude is bang-bang without singular arc, thus the optimal thrust ratio is determined by

$$u^*(t) = \begin{cases} 1, & S(t) \le 0, \\ 0, & S(t) > 0, \end{cases} \tag{9}$$

where $S(t)$ is the switching function defined as follows:

$$S(t) = 1 - \frac{T_m p_m(t)}{I_{\mathrm{sp}} g_0} - \frac{T_m}{m(t)} \|\boldsymbol{p}_v(t)\|_2. \tag{10}$$

The discontinuous bang-bang control often causes numerical difficulties, and the smoothing technique can be employed to solve this issue.[39–42] Here, we adopt the smoothing technique in Ref. 42 to approximate Eq. (9), i.e.

$$u^*(t) \approx u^*(t, \delta) = \frac{1}{2}\left(1 - \frac{S(t)}{\sqrt{\delta + |S(t)|^2}}\right), \tag{11}$$

where δ is the smoothing constant. Since the final mass is free, the transversality condition implies

$$p_m(t_f) = 0. \tag{12}$$

In addition, since the Hamiltonian in Eq. (6) does not contain time explicitly and the final time is free, the stationary condition states that

$$\mathscr{H}(t_f) = 0. \tag{13}$$

Equations (1), (7), (8), and (11) constitute a set of ordinary differential equations in terms of the states and the co-states. Given the initial condition described in Eq. (2) and co-states $\boldsymbol{p}_{r_0}$, $\boldsymbol{p}_{v_0}$, p_{m_0} and the final time t_f to be determined, the final states and co-states can be calculated by integrating Eqs. (1) and (7) along with the optimal control equations provided in Eqs. (8) and (11). In this way, the OCP is transformed into a TPBVP, which seeks to find the unknowns $\boldsymbol{p}_{r_0}$, $\boldsymbol{p}_{v_0}$, p_{m_0} and t_f so that the final condition specified in Eq. (3), the transversality condition in Eq. (12) and the stationary condition in Eq. (13) are met. The unknowns can be found by solving the following shooting function $\boldsymbol{\psi}$ defined as follows:

$$\boldsymbol{\psi}(\boldsymbol{p}_{r_0}; \boldsymbol{p}_{v_0}; p_{m_0}; t_f) = [\boldsymbol{r}(t_f); \boldsymbol{v}(t_f); p_m(t_f); \mathscr{H}(t_f)], \tag{14}$$

where $\boldsymbol{p}_{r_0}$, $\boldsymbol{p}_{v_0}$, and t_f are the initial guesses of $\boldsymbol{p}_r$, $\boldsymbol{p}_v$, and the final time, respectively.

3.2. *With the final steering angle constraint*

Solving Eq. (14) will provide only fuel optimal solution with the constraint in Eq. (4) unsatisfied. To this end, we modify the cost functional in Eq. (5) by augmenting a regularization term defined as follows:

$$\Delta(z,\theta,\beta,\epsilon) := \frac{1}{2}\exp(\beta z)\frac{\theta^2}{z+\epsilon}, \tag{15}$$

where β is a small negative constant and ϵ is a very small positive constant to avoid numerical singularity. In this way, the augmented cost functional $\tilde{J}$ is

$$\tilde{J} = \int_0^{\tilde{t}_f}(1+\Delta)u(\tilde{t})\mathrm{d}\tilde{t} = \int_0^{\tilde{t}_f}\left\{1+\frac{1}{2}\exp[\beta\tilde{z}(\tilde{t})]\frac{\tilde{\theta}^2(\tilde{t})}{\tilde{z}(\tilde{t})+\epsilon}\right\}\tilde{u}(\tilde{t})\mathrm{d}\tilde{t}, \tag{16}$$

where the symbol $\tilde{}$ will be used in the context of the augmented cost functional hereafter.

Now, we shall show that the following property holds for the regularization term $\Delta(\tilde{z}(\tilde{t}),\tilde{\theta}(\tilde{t}),\beta,\epsilon)$:

Property: The term $\Delta(\tilde{z}(\tilde{t}),\tilde{\theta}(\tilde{t}),\beta,\epsilon)$ remains nonnegative and sufficiently small for $\tilde{t}\in[0,\tilde{t}_f]$.

Proof. Note that $\tilde{z}(\tilde{t})>0$ holds before touchdown and β is a small negative constant, then the term $\exp[\beta\tilde{z}(\tilde{t})]$ becomes a very small positive number. It is clear that $\frac{\tilde{\theta}^2(\tilde{t})}{\tilde{z}(\tilde{t})+\epsilon}\geq 0$ always holds during the entire landing. To minimize the cost functional $\tilde{J}$, the nonnegative term $\frac{\tilde{\theta}^2(\tilde{t})}{\tilde{z}(\tilde{t})+\epsilon}$ cannot approach infinity. At touchdown $\tilde{t}=\tilde{t}_f$, we have $\tilde{z}(\tilde{t})=0$. In this case, both $\lim_{\tilde{t}\to\tilde{t}_f}\tilde{\theta}(\tilde{t})=0$ and $\lim_{\tilde{t}\to\tilde{t}_f}\frac{\tilde{\theta}^2(\tilde{t})}{\tilde{z}(\tilde{t})+\epsilon}=0$ have to hold in order for the term $\Delta(\tilde{z}(\tilde{t}),\tilde{\theta}(\tilde{t}),\beta,\epsilon)$ not to blow up. This completes the proof. □

Thus, the final steering angle constraint in Eq. (4) will be met automatically without causing too much penalty on the original cost functional in Eq. (5). In what follows, we will derive the corresponding TPBVP concerning the augmented cost functional $\tilde{J}$.

Note that the dynamics of the lunar lander remains the same as in Eq. (1), i.e.

$$\dot{\tilde{\boldsymbol{x}}}(\tilde{t}) = \boldsymbol{f}(\tilde{\boldsymbol{x}},\tilde{u},\tilde{\theta},\tilde{t}). \tag{17}$$

Then, the augmented Hamiltonian, denoted by $\tilde{\mathscr{H}}$, is expressed as follows:

$$\tilde{\mathscr{H}} = \tilde{\boldsymbol{p}}_r^T\cdot\tilde{\boldsymbol{v}} + \tilde{\boldsymbol{p}}_v^T\cdot\left(\boldsymbol{g}+\frac{\tilde{u}T_{\mathrm{m}}}{\tilde{m}}\hat{\boldsymbol{i}}_{\tilde{\theta}}\right)+\tilde{p}_m\left(-\frac{\tilde{u}T_{\mathrm{m}}}{I_{\mathrm{sp}}g_0}\right)+\left[1+\frac{1}{2}\exp(\beta\tilde{z})\frac{\tilde{\theta}^2}{\tilde{z}+\epsilon}\right]\tilde{u}. \tag{18}$$

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The dynamics of co-states is rebuilt as follows:

$$\begin{cases} \dot{\tilde{p}}_y(\tilde{t}) = -\dfrac{\partial \tilde{\mathscr{H}}}{\partial \tilde{y}(\tilde{t})} = 0, \\ \dot{\tilde{p}}_z(\tilde{t}) = -\dfrac{\partial \tilde{\mathscr{H}}}{\partial \tilde{z}(\tilde{t})} = -\dfrac{1}{2}\,\tilde{u}(\tilde{t})\tilde{\theta}^2(\tilde{t})\exp[\beta\tilde{z}(\tilde{t})]\,\dfrac{\beta[\tilde{z}(\tilde{t})+\epsilon]-1}{[\tilde{z}(\tilde{t})+\epsilon]^2}, \\ \dot{\tilde{\boldsymbol{p}}}_v(\tilde{t}) = -\dfrac{\partial \tilde{\mathscr{H}}}{\partial \tilde{\boldsymbol{v}}(\tilde{t})} = -\tilde{\boldsymbol{p}}_r(\tilde{t}), \\ \dot{\tilde{p}}_m(\tilde{t}) = -\dfrac{\partial \tilde{\mathscr{H}}}{\partial \tilde{m}(\tilde{t})} = \dfrac{\tilde{u}(\tilde{t})T_m}{\tilde{m}^2(\tilde{t})}\tilde{\boldsymbol{p}}_v^T(\tilde{t})\cdot\hat{\boldsymbol{i}}_{\tilde{\theta}}(\tilde{t}). \end{cases} \tag{19}$$

The optimal steering angle follows that

$$\frac{\partial \tilde{\mathscr{H}}}{\partial \tilde{\theta}(\tilde{t})} = 0. \tag{20}$$

Explicitly rewriting Eq. (20) leads to

$$\frac{\tilde{u}(\tilde{t})T_m}{\tilde{m}(\tilde{t})}[\tilde{p}_{v_y}(\tilde{t})\cos\tilde{\theta}(\tilde{t}) - \tilde{p}_{v_z}(\tilde{t})\sin\tilde{\theta}(\tilde{t})] + \tilde{u}(\tilde{t})\exp[\beta\tilde{z}(\tilde{t})]\frac{\tilde{\theta}(\tilde{t})}{\tilde{z}(\tilde{t})+\epsilon} = 0. \tag{21}$$

Since $\tilde{u}(\tilde{t}) \geq 0$ during landing, in order for Eq. (21) to hold for $\tilde{t} \in [0, \tilde{t}_f]$, we have

$$\frac{T_m}{\tilde{m}(\tilde{t})}[\tilde{p}_{v_y}(\tilde{t})\cos\tilde{\theta}^*(\tilde{t}) - \tilde{p}_{v_z}(\tilde{t})\sin\tilde{\theta}^*(\tilde{t})] + \exp[\beta\tilde{z}(\tilde{t})]\frac{\tilde{\theta}^*(\tilde{t})}{\tilde{z}(\tilde{t})+\epsilon} = 0. \tag{22}$$

Unlike the analytical solution in terms of $\boldsymbol{p}_v(t)$ in Eq. (8), the optimal steering angle considered here has to be determined by solving Eq. (22). However, Eq. (22) is a transcendental equation that may have multiple zeros. Thus, the typical Newton-like iterative method or bisection method may not find the desired zero. Based on the transforming procedure proposed by Zheng *et al.* in Ref. 43, we now show how to find the right zero for Eq. (22).

Differentiating Eq. (22) w.r.t. $\tilde{\theta}$ leads to

$$\frac{T_m}{\tilde{m}}(-\tilde{p}_{v_y}\sin\tilde{\theta}^* - \tilde{p}_{v_z}\cos\tilde{\theta}^*) + \exp(\beta\tilde{z})\frac{1}{\tilde{z}+\epsilon} = 0. \tag{23}$$

By substituting the half-angle formulas

$$\sin\tilde{\theta} = \frac{2\tan\frac{\tilde{\theta}}{2}}{1+\tan^2\frac{\tilde{\theta}}{2}} \quad \text{and} \quad \cos\tilde{\theta} = \frac{1-\tan^2\frac{\tilde{\theta}}{2}}{1+\tan^2\frac{\tilde{\theta}}{2}},$$

into Eq. (23), one has that $\tan\frac{\tilde{\theta}}{2}$ is a zero of the following quadratic polynomial in terms of x:

$$\left[-\tilde{p}_{v_z} - \frac{\tilde{m}\exp(\beta\tilde{z})}{T_m(\tilde{z}+\epsilon)}\right]x^2 + 2\tilde{p}_{v_y}x + \tilde{p}_{v_z} - \frac{\tilde{m}\exp(\beta\tilde{z})}{T_m(\tilde{z}+\epsilon)} = 0. \tag{24}$$

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It is evident that the two roots of Eq. (24) can be obtained either by radicals or standard polynomial solvers. A bisection method can be then used to find all the zeros of Eq. (22) by comparing the values for Eq. (22) when $\tilde{\theta}$ takes the values of $-\pi$, π, and the two roots of Eq. (24). After all the zeros of Eq. (22) are found, the optimal value for $\tilde{\theta}$ is whichever that minimizes the Hamiltonian in Eq. (18).

Since the thrust ratio is still linear w.r.t. both the state dynamics in Eq. (1) and the augmented cost functional in Eq. (16), it is reasonable to assume that the optimal thrust ratio is still in a bang-bang form, i.e.

$$\tilde{u}^*(\tilde{t}) = \begin{cases} 1, & \tilde{S}(\tilde{t}) \leq 0, \\ 0, & \tilde{S}(\tilde{t}) > 0, \end{cases} \tag{25}$$

where $\tilde{S}(\tilde{t})$ is defined as follows:

$$\tilde{S}(\tilde{t}) = \tilde{\boldsymbol{p}}_v^T \cdot \hat{\boldsymbol{i}}_{\tilde{\theta}} \frac{T_m}{\tilde{m}} - \frac{T_m}{I_{\text{sp}} g_0} \tilde{p}_m + 1 + \frac{1}{2} \exp(\beta \tilde{z}) \frac{\tilde{\theta}^2}{\tilde{z} + \epsilon}. \tag{26}$$

Likewise, the discontinuous bang-bang control in Eq. (25) is approximated by

$$\tilde{u}^*(\tilde{t}) \approx \tilde{u}^*(\tilde{t}, \delta) = \frac{1}{2} \left(1 - \frac{\tilde{S}(\tilde{t})}{\sqrt{\delta + |\tilde{S}(\tilde{t})|^2}} \right), \tag{27}$$

where δ is the same smoothing constant as in Eq. (11). The transversality condition concerning $\tilde{p}_m$ stays unchanged, i.e.

$$\tilde{p}_m(\tilde{t}_f) = 0. \tag{28}$$

Meanwhile, the stationary condition is

$$\tilde{\mathscr{H}}(\tilde{t}_f) = 0. \tag{29}$$

The initial condition in Eq. (2) and the final condition in Eq. (3) still apply. As a result, the corresponding TPBVP amounts to solving the following shooting function:

$$\boldsymbol{\psi}(\tilde{\boldsymbol{p}}_{r_0}; \tilde{\boldsymbol{p}}_{v_0}; \tilde{p}_{m_0}; \tilde{t}_f) = [\tilde{\boldsymbol{r}}(\tilde{t}_f); \tilde{\boldsymbol{v}}(\tilde{t}_f); \tilde{p}_m(\tilde{t}_f); \tilde{\mathscr{H}}(\tilde{t}_f)], \tag{30}$$

where $\tilde{\boldsymbol{p}}_{r_0}$, $\tilde{\boldsymbol{p}}_{v_0}$, $\tilde{p}_{m_0}$, and $\tilde{t}_f$ are the initial guesses of $\tilde{\boldsymbol{p}}_r$, $\tilde{\boldsymbol{p}}_v$, $\tilde{p}_m$, and the final time, respectively.

4. Numerical Simulations

Before proceeding, we shall present some constants for simulations. The propulsion system of the lunar lander is specified by $I_{\text{sp}} = 311\,\text{s}$ and $T_m = 44,000\,\text{N}$.

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The gravitational acceleration of the Moon is $g = 1.6229\ \mathrm{m/s^2}$, and g_e is equal to $9.81\,\mathrm{m/s^2}$. The absolute and relative tolerances for propagating the dynamics of states and co-states are both set as 1.0×10^{-10}. The termination tolerance for *f solve* is set as 1.0×10^{-10}. The smoothing constant δ is 1.0×10^{-10}.

In the following, we shall first demonstrate the impact of different parameter settings in Eq. (16) on the landing trajectory. Then, we consider two quite different initial conditions to illustrate the efficacy of the proposed method. Lastly, we test the applicability of the proposed method to various initial conditions.

4.1. *Impact of the parameter setting*

Unlike the work in Ref. 33, where the modification term needs the iteration procedure, the proposed method only has two constant parameters. Consider an initial condition of the lunar lander set as follows:

$$\boldsymbol{x}_0 = [-80\,\mathrm{m}, 200\,\mathrm{m}, 20\,\mathrm{m/s}, -30\,\mathrm{m/s}, 9400\,\mathrm{kg}].$$

We first solve Eq. (14) to obtain the solution without the final steering angle constraint. The steering angle and the mass profiles are shown in Figs. 2 and 3,

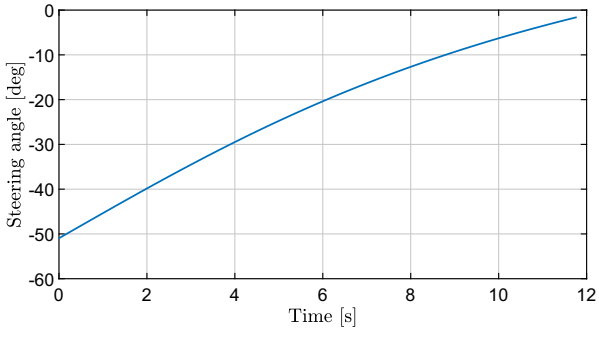


Fig. 2. Steering angle profile during landing.

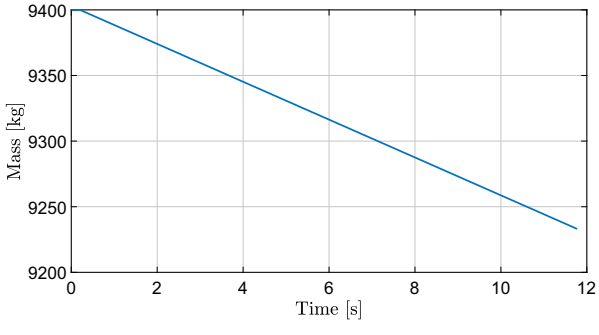


Fig. 3. Mass profile during landing.

respectively. It can be seen that the final steering angle is not zero. As a result, the optimal fuel consumption is 166.9223 kg.

Table 1 lists six parameter settings for the two parameters β and ϵ in Eq. (16). After using the six parameter settings, the corresponding final steering angle and the penalty on fuel consumption are compared in Table 2. As shown by Table 2, the final steering angle becomes closer to zero as the negative parameter β increases for a given ϵ. On the other hand, the penalty on the fuel consumption becomes smaller as the parameter ϵ decreases. As a result, both parameters have influence on the solution. Since the parameter setting $\beta = -1.0 \times 10^{-2}$ and $\epsilon = 1.0 \times 10^{-8}$ incurs only negligible penalty on the fuel consumption and the final steering angle error is very small, we will use this parameter setting hereafter.

Table 1. Six different parameter settings.

Parameter	β	ϵ
Setting 1	-1.0×10^{-2}	1.0×10^{-8}
Setting 2	-1.0×10^{-2}	1.0×10^{-6}
Setting 3	-1.0×10^{-2}	1.0×10^{-4}
Setting 4	-1.0×10^{-1}	1.0×10^{-8}
Setting 5	-1.0×10^{-1}	1.0×10^{-6}
Setting 6	-1.0×10^{-1}	1.0×10^{-4}

Table 2. Comparison of the final steering angle and the penalty on the fuel consumption with different parameter settings.

Item	Final steering angle (deg)	Penalty on the fuel consumption (kg)
Setting 1	-4.1798×10^{-8}	0.0092
Setting 2	-4.1436×10^{-6}	0.0093
Setting 3	-4.1429×10^{-4}	0.0093
Setting 4	-3.6590×10^{-8}	0.0101
Setting 5	-3.6579×10^{-6}	0.0101
Setting 6	-3.6563×10^{-4}	0.0101

4.2. *Efficacy of the proposed method*

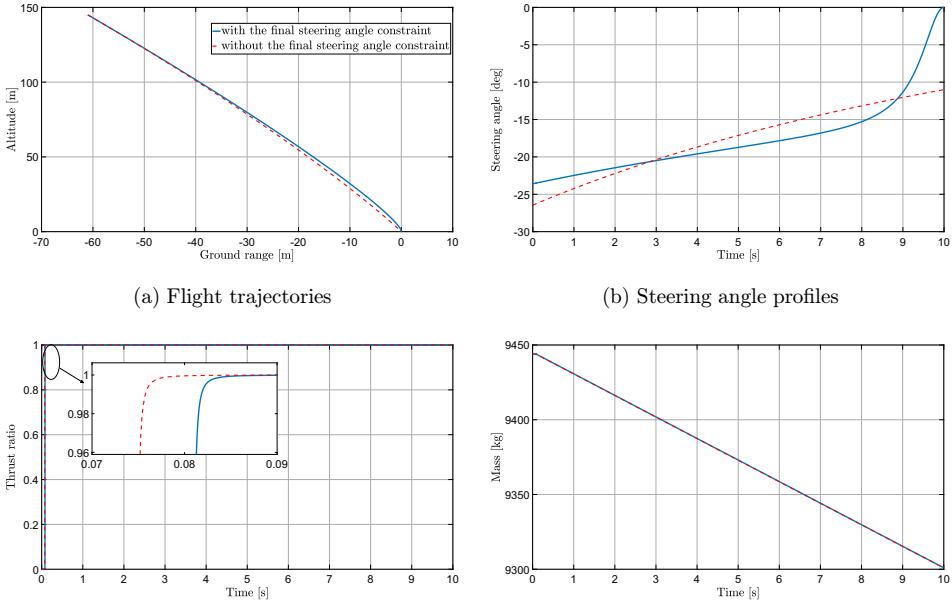
Consider two different initial conditions, which are listed in Table 3. Regarding Case A, solving the shooting function in Eqs. (14) and (30) enables one to obtain the optimal solution without and with the final steering angle constraint. Figure 4 displays

Table 3. Two different initial conditions.

Parameter	y_0 (m)	z_0 (m)	v_{y_0} (m/s)	v_{z_0} (m/s)	m_0 (kg)
Case A	−61	145	14	−28	9444
Case B	1207.7454	7639.0205	−29.7321	−210.9858	10018.3944

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(a) Flight trajectories

(b) Steering angle profiles

(c) Thrust ratio profiles

(d) Mass profiles

Fig. 4. Trajectories, steering angle, thrust ratio, and mass profiles with and without the final steering angle constraint for Case A.

the flight trajectories, steering angle, and thrust ratio, as well as mass profiles during landing. As seen from Figs. 4(a) and 4(b), the flight trajectories are different, in which the final steering angle obtained from Eq. (30) reaches zero as expected; in contrast, the steering angle obtained from Eq. (14) is progressing linearly and arrives at -11.0205° at touchdown. As for the thrust ratio, both solutions show an "off-on" structure. However, the switching time is slightly different; the engine is switched from off to on at $\tilde{t} = 0.0811$ s for the solution with the final steering angle constraint. On the other hand, for the solution without the final steering angle constraint, the engine is switched on at $t = 0.0748$ s, as shown by the scaled plot in Fig. 4(c). As a result, the final time for the solution with and without the final steering angle constraint is $\tilde{t}_f = 9.9994$ s and $t_f = 9.9779$ s, respectively. Moreover, the final mass for these two solutions is $\tilde{m}(\tilde{t}_f) = 9300.9654$ kg and $m(t_f) = 9301.1852$ kg, respectively. This indicates that adding the regularization term in Eq. (15) only results in an extra fuel consumption of 0.2198 kg, which is negligible in the context of an initial mass of 9444 kg.

In addition, Fig. 5 shows the regularization term profile defined in Eq. (15), from which we can see that the regularization term Δ remains nonnegative and small during the entire landing. Specifically, before touchdown, Δ is always a small positive number; while at touchdown, $\Delta(\tilde{t}_f)$ becomes zero because the final steering angle $\tilde{\theta}(\tilde{t}_f)$ becomes zero. Figure 6 compares some co-states and the Hamiltonian for these

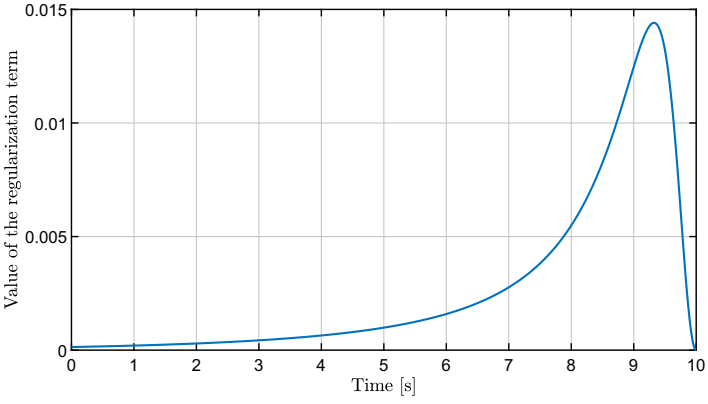
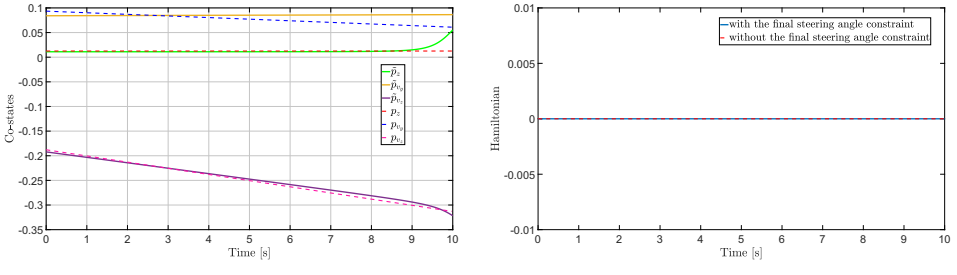


Fig. 5. Regularization term profile during landing.

two solutions. It is clear from Fig. 6 that p_z is constant, while p_{v_y} and p_{v_z} change linearly w.r.t time, as predicted by Eq. (7). On the other hand, for the solution with the final steering angle constraint, both $\tilde{p}_{v_y}$ and $\tilde{p}_{v_z}$ are progressing nonlinearly w.r.t time. Moreover, the optimality of the obtained solutions can be further verified by the Hamiltonian in Fig. 6(b), where both solutions remain zero during the entire landing.



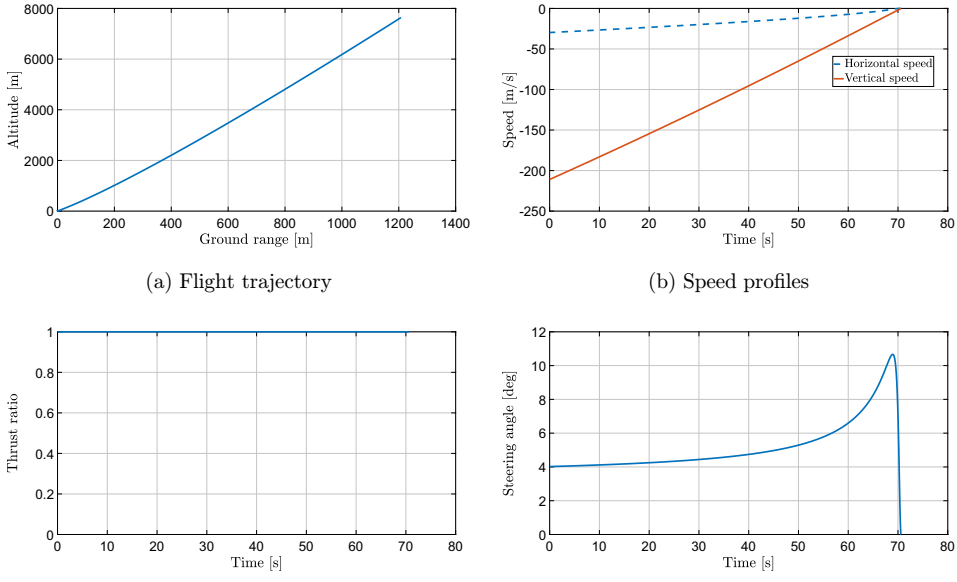
(a) Comparison of some co-states

(b) Comparison of the Hamiltonian

Fig. 6. Comparison of some co-states and the Hamiltonian with and without the final steering angle constraint for Case A.

We now move to Case B. Since the efficacy is demonstrated by Case B, we directly give the results, as shown in Fig. 7. Different from Case A, the initial altitude is much greater, leading to a much longer landing time. It can be seen from Figs. 7(a) and 7(b) that the final condition in Eq. (3) is rigorously satisfied. As for the thrust ratio, the optimal thrust magnitude is kept at its maximum during the entire landing. In addition, the steering angle experiences slow change before the final phase of the landing, and it rapidly changes and convergences to zero, leading to the satisfaction in the final steering angle constraint in Eq. (4).

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(a) Flight trajectory

(b) Speed profiles

(c) Thrust ratio profile

(d) Steering angle profile

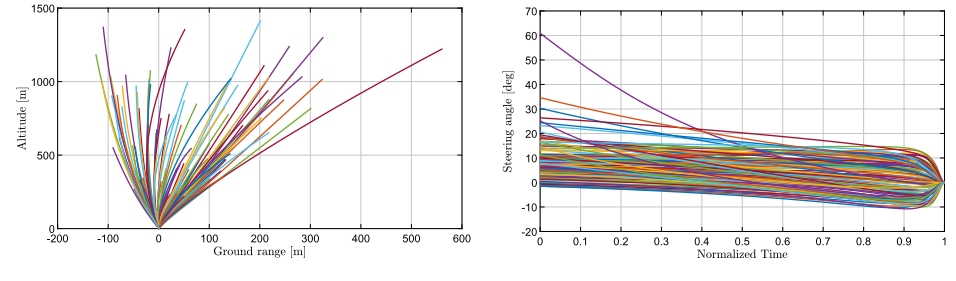
Fig. 7. Flight trajectory, speed, thrust ratio, and steering angle profiles with the final steering angle constraint for Case B.

4.3. *Applicability of the proposed method*

To further demonstrate the applicability of the proposed method, a total of 100 initial conditions are randomly generated in a domain $\mathcal{A}$, defined by

$$y_0 \in [-125, 600]\ (\mathrm{m}), \quad z_0 \in [50, 1500]\ (\mathrm{m}),$$
$$v_{y_0} \in [-50, 10]\ (\mathrm{m/s}), \quad v_{z_0} \in [-100, 10]\ (\mathrm{m/s}), \quad m_0 \in [9050, 9450]\ (\mathrm{kg}).$$

After solving Eq. (30) regarding these 100 initial conditions, Fig. 8 shows the flight trajectories, steering angle, horizontal speed, and vertical speed profiles. It can be

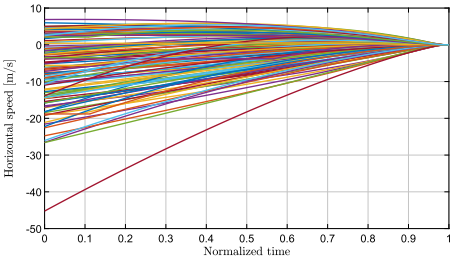


(a) Flight trajectories

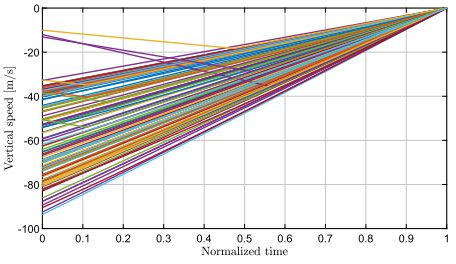
(b) Steering angle profiles

Fig. 8. Trajectories, steering angle, horizontal speed, and vertical speed profiles with the final steering angle constraint.

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(c) Horizontal speed profiles



(d) Vertical speed profiles

Fig. 8. (*Continued*)

observed that vertical landing is achieved for all cases, which can be further verified by the fact that the final steering angle reaches zero, as shown in Fig. 8(b). From Figs. 8(c) and 8(d), we can see that the horizontal speed and vertical speed become zero at touchdown. Furthermore, as for the vertical speed in most cases, it is monotonically decreasing to zero because the engine thrust is kept at the maximum during the entire landing. For these cases where the vertical speed is initially increasing solely due to gravity, the engine is switched off during this time interval. Then, the engine is kept on until touchdown.

5. Conclusion

In this work we proposed an easy-to-implement method to generate the fuel-optimal vertical landing trajectory for a lunar lander. To ensure the vertical landing, we modified the cost functional by augmenting a nonnegative regularization term. In this way, in order for the Hamiltonian to be minimized, the final steering angle would be satisfied automatically. By employing a transforming procedure, the necessary conditions for optimality indicated that the modified optimal steering angle could be found by a simple bisection method. As a result, the fuel-optimal vertical landing trajectory could be generated offline by the indirect shooting method. Future research directions include generalization of the proposed method to vertical landing in three dimension and its extension to onboard application.

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